

R Code

```
1 library(data.table)
2 library(ggplot2)
3 library(tidyr)
4
5 data <- read_csv("C:/Users/hites/OneDrive/Desktop/Hitesh/Portfolio/Additional
Skills/Data Analytics Job Simulation - Quantum/data/QVI_data.csv")
6 View(data)
7 setDT(data)
8 theme_set(theme_bw())
9 theme_update(plot.title = element_text(hjust = 0.5))
10
11 # Create YEARMONTH
12 data[, DATE := as.Date(DATE)]
13 data[, YEARMONTH := year(DATE) * 100 + month(DATE)]
14
15 class(data)
16 names(data)
17 head(data)
18
19 #Create monthly store metrics
20
21 measureOverTime <- data[, .(
22   totSales = sum(TOT_SALES),
23   nCustomers = uniqueN(LYLTY_CARD_NBR),
24   nTxnPerCust = uniqueN(TXN_ID) / uniqueN(LYLTY_CARD_NBR),
25   nChipsPerTxn = sum(PROD_QTY) / uniqueN(TXN_ID),
26   avgPricePerUnit = sum(TOT_SALES) / sum(PROD_QTY)
27 ), by = .(STORE_NBR, YEARMONTH)][order(STORE_NBR, YEARMONTH)]
28
29 head(measureOverTime)
30 summary(measureOverTime)
31
32 # Filter stores with full 12 months
33 storesWithFullObs <- unique(
34   measureOverTime[, .N, STORE_NBR][N == 12, STORE_NBR]
35 )
36
37 preTrialMeasures <- measureOverTime[
38   YEARMONTH < 201902 & STORE_NBR %in% storesWithFullObs,
39 ]
40 length(storesWithFullObs)
41 head(preTrialMeasures)
42
43 # Create the correlation function
44
45 calculateCorrelation <- function(inputTable, metricCol, storeComparison) {
46
47   calcCorrTable <- data.table(
48     Store1 = numeric(),
49     Store2 = numeric(),
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50     corr_measure = numeric()
51 )
52
53 storeNumbers <- unique(inputTable[, STORE_NBR])
54
55 for (i in storeNumbers) {
56
57     calculatedMeasure <- data.table(
58         Store1 = storeComparison,
59         Store2 = i,
60         corr_measure = cor(
61             inputTable[STORE_NBR == storeComparison, eval(metricCol)],
62             inputTable[STORE_NBR == i, eval(metricCol)]
63         )
64     )
65
66     calcCorrTable <- rbind(calcCorrTable, calculatedMeasure)
67 }
68
69 return(calcCorrTable)
70 }
71
72 calculateMagnitudeDistance <- function(inputTable, metricCol, storeComparison) {
73
74     calcDistTable <- data.table(
75         Store1 = numeric(),
76         Store2 = numeric(),
77         YEARMONTH = numeric(),
78         measure = numeric()
79     )
80
81     storeNumbers <- unique(inputTable[, STORE_NBR])
82
83     for (i in storeNumbers) {
84
85         calculatedMeasure <- data.table(
86             Store1 = storeComparison,
87             Store2 = i,
88             YEARMONTH = inputTable[STORE_NBR == storeComparison, YEARMONTH],
89             measure = abs(
90                 inputTable[STORE_NBR == storeComparison, eval(metricCol)] -
91                 inputTable[STORE_NBR == i, eval(metricCol)]
92             )
93         )
94
95         calcDistTable <- rbind(calcDistTable, calculatedMeasure)
96     }
97
98     minMaxDist <- calcDistTable[, .(
99         minDist = min(measure),
100         maxDist = max(measure)
101     ), by = .(Store1, YEARMONTH)]
102
103     distTable <- merge(calcDistTable, minMaxDist, by = c("Store1", "YEARMONTH"))

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104
105     distTable[, magnitudeMeasure := 1 - (measure - minDist) / (maxDist - minDist)]
106
107     finalDistTable <- distTable[, .(
108         mag_measure = mean(magnitudeMeasure)
109     ), by = .(Store1, Store2)]
110
111     return(finalDistTable)
112 }
113
114 # Find Control Store for Trial Store 77
115
116 trial_store <- 77
117
118 corr_nSales <- calculateCorrelation(preTrialMeasures, quote(totSales), trial_store)
119 corr_nCustomers <- calculateCorrelation(preTrialMeasures, quote(nCustomers),
120 trial_store)
121
122 magnitude_nSales <- calculateMagnitudeDistance(preTrialMeasures, quote(totSales),
123 trial_store)
124 magnitude_nCustomers <- calculateMagnitudeDistance(preTrialMeasures,
125 quote(nCustomers), trial_store)
126
127 corr_weight <- 0.5
128
129 score_nSales <- merge(
130     corr_nSales,
131     magnitude_nSales,
132     by = c("Store1", "Store2")
133 )[, scoreNSales := corr_weight * corr_measure + (1 - corr_weight) * mag_measure]
134
135 score_nCustomers <- merge(
136     corr_nCustomers,
137     magnitude_nCustomers,
138     by = c("Store1", "Store2")
139 )[, scoreNCust := corr_weight * corr_measure + (1 - corr_weight) * mag_measure]
140
141 score_Control <- merge(
142     score_nSales[, .(Store1, Store2, scoreNSales)],
143     score_nCustomers[, .(Store1, Store2, scoreNCust)],
144     by = c("Store1", "Store2")
145 )
146
147 score_Control[, finalControlScore := 0.5 * scoreNSales + 0.5 * scoreNCust]
148
149 score_Control[order(-finalControlScore)][1:10]
150 # Control Store Selection - Trial Store 77
151 # Store 233 achieved the highest combined score based on
152 # sales and customer metrics during the pre-trial period.
153 # Therefore Store 233 was selected as the control store
154 # for Trial Store 77.
155
156 # Visual Validation
157 pastSales <- measureOverTime[

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155     STORE_NBR %in% c(77,233)
156 ]
157
158 ggplot(
159   pastSales,
160   aes(
161     x = YEARMONTH,
162     y = totSales,
163     color = factor(STORE_NBR)
164   )
165 ) +
166   geom_line(size = 1) +
167   geom_point(size = 2) +
168   labs(
169     title = "Monthly Total Sales: Trial Store 77 vs Control Store 233",
170     x = "Year Month",
171     y = "Total Sales",
172     color = "Store"
173   )
174
175 # Customer Chart Comparison
176
177 ggplot(
178   pastSales,
179   aes(
180     x = YEARMONTH,
181     y = nCustomers,
182     color = factor(STORE_NBR)
183   )
184 ) +
185   geom_line(size = 1) +
186   geom_point(size = 2) +
187   labs(
188     title = "Monthly Customers: Trial Store 77 vs Control Store 233",
189     x = "Year Month",
190     y = "Customers",
191     color = "Store"
192   )
193
194 # Find Control Store for Trial Store 86
195
196 trial_store <- 86
197
198 corr_nSales <- calculateCorrelation(preTrialMeasures, quote(totSales), trial_store)
199 corr_nCustomers <- calculateCorrelation(preTrialMeasures, quote(nCustomers),
200   trial_store)
201
202 magnitude_nSales <- calculateMagnitudeDistance(preTrialMeasures, quote(totSales),
203   trial_store)
204 magnitude_nCustomers <- calculateMagnitudeDistance(preTrialMeasures,
205   quote(nCustomers), trial_store)

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206 score_nSales <- merge(
207   corr_nSales,
208   magnitude_nSales,
209   by = c("Store1", "Store2")
210 )[, scoreNSales := corr_weight * corr_measure + (1 - corr_weight) * mag_measure]
211
212 score_nCustomers <- merge(
213   corr_nCustomers,
214   magnitude_nCustomers,
215   by = c("Store1", "Store2")
216 )[, scoreNCust := corr_weight * corr_measure + (1 - corr_weight) * mag_measure]
217
218 score_Control <- merge(
219   score_nSales[, .(Store1, Store2, scoreNSales)],
220   score_nCustomers[, .(Store1, Store2, scoreNCust)],
221   by = c("Store1", "Store2")
222 )
223
224 score_Control[, finalControlScore := 0.5 * scoreNSales + 0.5 * scoreNCust]
225
226 score_Control[order(-finalControlScore)][1:10]
227
228 # Control Store Selection - Trial Store 86
229 # Store 155 achieved the highest combined score based on
230 # sales and customer metrics during the pre-trial period.
231 # Therefore Store 155 was selected as the control store
232 # for Trial Store 86.
233
234 # Trial Store 88
235
236 trial_store <- 88
237
238 corr_nSales <- calculateCorrelation(preTrialMeasures, quote(totSales), trial_store)
239 corr_nCustomers <- calculateCorrelation(preTrialMeasures, quote(nCustomers),
240   trial_store)
241
242 magnitude_nSales <- calculateMagnitudeDistance(preTrialMeasures, quote(totSales),
243   trial_store)
244 magnitude_nCustomers <- calculateMagnitudeDistance(preTrialMeasures,
245   quote(nCustomers), trial_store)
246
247 score_nSales <- merge(
248   corr_nSales,
249   magnitude_nSales,
250   by = c("Store1", "Store2")
251 )[, scoreNSales := 0.5 * corr_measure + 0.5 * mag_measure]
252
253 score_nCustomers <- merge(
254   corr_nCustomers,
255   magnitude_nCustomers,
256   by = c("Store1", "Store2")
257 )[, scoreNCust := 0.5 * corr_measure + 0.5 * mag_measure]
258
259 score_Control <- merge(

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257     score_nSales[, .(Store1, Store2, scoreNSales)],
258     score_nCustomers[, .(Store1, Store2, scoreNCust)],
259     by = c("Store1", "Store2")
260 )
261
262 score_Control[, finalControlScore := 0.5 * scoreNSales + 0.5 * scoreNCust]
263
264 score_Control[order(-finalControlScore)][1:10]
265
266
267 # Control Store Selection Results
268 #
269 # Trial Store 77 → Control Store 233
270 # Trial Store 86 → Control Store 155
271 # Trial Store 88 → Control Store 237
272 #
273 # Control stores were selected based on a combined score
274 # incorporating both Pearson correlation and magnitude distance
275 # for total sales and number of customers during the pre-trial period.
276 #
277 # These stores demonstrated the strongest similarity to the
278 # respective trial stores before the trial commenced.
279
280
281 # Create a Reusable Plot Function
282
283 plotTrialControl <- function(store1, store2, metric) {
284
285     plotData <- measureOverTime[
286         STORE_NBR %in% c(store1, store2)
287     ]
288
289     ggplot(
290         plotData,
291         aes(
292             x = YEARMONTH,
293             y = get(metric),
294             colour = factor(STORE_NBR),
295             group = STORE_NBR
296         )
297     ) +
298     geom_line(size = 1) +
299     geom_point(size = 2) +
300     labs(
301         title = paste(metric, "-", store1, "vs", store2),
302         colour = "Store"
303     )
304 }
305
306 # Set Trial and Control Store
307 trial_store <- 77
308 control_store <- 233
309
310

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311 # Visual Check - Total Sales
312 measureOverTimeSales <- copy(measureOverTime)
313
314 pastSales <- measureOverTimeSales[
315   , Store_type := ifelse(
316     STORE_NBR == trial_store,
317     "Trial",
318     ifelse(STORE_NBR == control_store, "Control", "Other stores")
319   )
320 ][
321   , totSales := mean(totSales),
322   by = c("YEARMONTH", "Store_type")
323 ][
324   , TransactionMonth := as.Date(
325     paste(YEARMONTH %/% 100, YEARMONTH %% 100, 1, sep = "-"),
326     "%Y-%m-%d"
327   )
328 ][
329   YEARMONTH < 201903,
330 ]
331
332 ggplot(pastSales, aes(TransactionMonth, totSales, color = Store_type)) +
333   geom_line() +
334   labs(
335     x = "Month of operation",
336     y = "Total sales",
337     title = "Total sales by month"
338   )
339
340 # Visual check – number of customers
341
342 measureOverTimeCusts <- copy(measureOverTime)
343
344 pastCustomers <- measureOverTimeCusts[
345   , Store_type := ifelse(
346     STORE_NBR == trial_store,
347     "Trial",
348     ifelse(STORE_NBR == control_store, "Control", "Other stores")
349   )
350 ][
351   , numberCustomers := mean(nCustomers),
352   by = c("YEARMONTH", "Store_type")
353 ][
354   , TransactionMonth := as.Date(
355     paste(YEARMONTH %/% 100, YEARMONTH %% 100, 1, sep = "-"),
356     "%Y-%m-%d"
357   )
358 ][
359   YEARMONTH < 201903,
360 ]
361
362 ggplot(pastCustomers, aes(TransactionMonth, numberCustomers, color = Store_type)) +
363   geom_line() +
364   labs(

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365     x = "Month of operation",
366     y = "Number of customers",
367     title = "Number of customers by month"
368 )
369
370 # Scale control store sale
371
372 scalingFactorForControlSales <- preTrialMeasures[
373   STORE_NBR == trial_store & YEARMONTH < 201902,
374   sum(totSales)
375 ] / preTrialMeasures[
376   STORE_NBR == control_store & YEARMONTH < 201902,
377   sum(totSales)
378 ]
379
380 scalingFactorForControlSales
381
382 # Apply scaling factor
383
384 measureOverTimeSales <- copy(measureOverTime)
385
386 scaledControlSales <- measureOverTimeSales[
387   STORE_NBR == control_store,
388 ][
389   , controlSales := totSales * scalingFactorForControlSales
390 ]
391
392 # Calculate percentage difference
393 percentageDiff <- merge(
394   scaledControlSales[, .(YEARMONTH, controlSales)],
395   measureOverTime[
396     STORE_NBR == trial_store,
397     .(YEARMONTH, trialSales = totSales)
398   ],
399   by = "YEARMONTH"
400 )[
401   , percentageDiff := abs(controlSales - trialSales) / controlSales
402 ]
403
404 percentageDiff
405
406 # Calculate Standard Deviation
407
408 stdDev <- sd(percentDiff[YEARMONTH < 201902, percentageDiff])
409
410 degreesOfFreedom <- 7
411
412 stdDev
413
414 # Calculate t values
415
416 percentDiff[
417   , tValue := percentDiff / stdDev
418 ]

```

```

419     , TransactionMonth := as.Date(
420       paste(YEARMONTH %/% 100, YEARMONTH %% 100, 1, sep = "-"),
421       "%Y-%m-%d"
422     )
423   ]
424
425   percentageDiff[
426     YEARMONTH >= 201902 & YEARMONTH <= 201904,
427     .(YEARMONTH, trialSales, controlSales, percentageDiff, tValue)
428   ]
429
430   # get Critical t value
431   qt(0.95, df = degreesOfFreedom)
432
433   # Insights:
434   # Trial Store 77 recorded significantly higher sales during
435   # March and April 2019 compared to the scaled control store.
436   #
437   # The t-values for March (7.34) and April (12.48) exceeded
438   # the critical t-value of 1.89, indicating a statistically
439   # significant increase in sales during the trial period.
440   #
441   # February 2019 did not show a statistically significant uplift.
442   #
443   # Overall, the trial appears to have had a positive impact on sales
444   # performance in Store 77.
445
446   # Create Trial Assessment charts
447
448   measureOverTimeSales <- copy(measureOverTime)
449
450   pastSales <- measureOverTimeSales[
451     , Store_type := ifelse(
452       STORE_NBR == trial_store,
453       "Trial",
454       ifelse(STORE_NBR == control_store, "Control", "Other stores")
455     )
456   ][
457     , totSales := mean(totSales),
458     by = c("YEARMONTH", "Store_type")
459   ][
460     , TransactionMonth := as.Date(
461       paste(YEARMONTH %/% 100, YEARMONTH %% 100, 1, sep = "-"),
462       "%Y-%m-%d"
463     )
464   ][
465     Store_type %in% c("Trial", "Control"),
466   ]
467
468   pastSales_Controls95 <- pastSales[
469     Store_type == "Control",
470   ][
471     , totSales := totSales * (1 + stdDev * 2)
472   ][

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473     , Store_type := "Control 95th % confidence interval"
474 ]
475
476 pastSales_Controls5 <- pastSales[
477   Store_type == "Control",
478 ][
479   , totSales := totSales * (1 - stdDev * 2)
480 ][
481   , Store_type := "Control 5th % confidence interval"
482 ]
483
484 trialAssessment <- rbind(
485   pastSales,
486   pastSales_Controls95,
487   pastSales_Controls5
488 )
489
490 ggplot(trialAssessment, aes(TransactionMonth, totSales, color = Store_type)) +
491   geom_rect(
492     data = trialAssessment[YEARMONTH < 201905 & YEARMONTH > 201901],
493     aes(
494       xmin = min(TransactionMonth),
495       xmax = max(TransactionMonth),
496       ymin = 0,
497       ymax = Inf,
498       color = NULL
499     ),
500     show.legend = FALSE
501   ) +
502   geom_line() +
503   labs(
504     x = "Month of operation",
505     y = "Total sales",
506     title = "Total sales by month"
507   )
508
509
510 #Customer Analysis
511 scalingFactorForControlCustomers <- preTrialMeasures[
512   STORE_NBR == trial_store & YEARMONTH < 201902,
513   sum(nCustomers)
514 ] / preTrialMeasures[
515   STORE_NBR == control_store & YEARMONTH < 201902,
516   sum(nCustomers)
517 ]
518
519 scalingFactorForControlCustomers
520
521 # Scale Customer Counts
522 scaledControlCustomers <- measureOverTime[
523   STORE_NBR == control_store,
524 ][
525   , controlCustomers := nCustomers * scalingFactorForControlCustomers
526 ]

```

```

527
528 # Calculate percentage difference
529 customerDiff <- merge(
530   scaledControlCustomers[, .(YEARMONTH, controlCustomers)],
531   measureOverTime[
532     STORE_NBR == trial_store,
533     .(YEARMONTH, trialCustomers = nCustomers)
534   ],
535   by = "YEARMONTH"
536 )
537   , percentageDiff := abs(controlCustomers - trialCustomers) /
538     controlCustomers
539 ]
540
541 # Standard Deviation
542 stdDevCust <- sd(
543   customerDiff[
544     YEARMONTH < 201902,
545     percentageDiff
546   ]
547 )
548
549 # Customer t value
550 customerDiff[
551   , tValue := percentageDiff / stdDevCust
552 ]
553
554 customerDiff[
555   YEARMONTH >= 201902 & YEARMONTH <= 201904,
556   .(
557     YEARMONTH,
558     trialCustomers,
559     controlCustomers,
560     percentageDiff,
561     tValue
562   )
563 ]
564
565 # Insights:
566 # Customer numbers increased significantly during March and April 2019.
567 #
568 # The customer uplift aligns with the observed increase in sales,
569 # suggesting that the trial layout attracted additional customers
570 # into the store rather than simply increasing spend from existing
571 # customers.
572 #
573 # Store 77 demonstrates strong evidence that the trial layout
574 # positively impacted performance.
575
576
577 # Craeting Function
578 assessTrialStore <- function(trial_store, control_store, metric_name) {
579
580   # Step 1: Scale pre-trial control store to match trial store

```

```

581 scalingFactor <- preTrialMeasures[
582   STORE_NBR == trial_store & YEARMONTH < 201902,
583   sum(get(metric_name))
584 ] / preTrialMeasures[
585   STORE_NBR == control_store & YEARMONTH < 201902,
586   sum(get(metric_name))
587 ]
588
589 # Step 2: Apply scaling factor to control store
590 scaledControl <- measureOverTime[
591   STORE_NBR == control_store,
592 ][
593   , scaledControlMetric := get(metric_name) * scalingFactor
594 ]
595
596 # Step 3: Compare scaled control against trial store
597 percentageDiff <- merge(
598   scaledControl[, .(YEARMONTH, scaledControlMetric)],
599   measureOverTime[
600     STORE_NBR == trial_store,
601     .(YEARMONTH, trialMetric = get(metric_name))
602   ],
603   by = "YEARMONTH"
604 )[
605   , percentageDiff := abs(scaledControlMetric - trialMetric) /
606     scaledControlMetric
607 ]
608
609 # Step 4: Calculate standard deviation from pre-trial period
610 stdDev <- sd(
611   percentageDiff[
612     YEARMONTH < 201902,
613     percentageDiff
614   ]
615 )
616
617 degreesOfFreedom <- 7
618 criticalValue <- qt(0.95, df = degreesOfFreedom)
619
620 # Step 5: Calculate t-values
621 percentageDiff[
622   , tValue := percentageDiff / stdDev
623 ][
624   , significant := tValue > criticalValue
625 ]
626
627 # Step 6: Return trial period only
628 result <- percentageDiff[
629   YEARMONTH >= 201902 & YEARMONTH <= 201904,
630   .(
631     trial_store = trial_store,
632     control_store = control_store,
633     metric = metric_name,
634     YEARMONTH,

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635         trialMetric,
636         scaledControlMetric,
637         percentageDiff,
638         tValue,
639         criticalValue,
640         significant
641     )
642 ]
643
644     return(result)
645 }
646
647 # Store 86 and 155
648 assessTrialStore(86, 155, "totSales")
649 assessTrialStore(86, 155, "nCustomers")
650
651 # Store 88 and 237
652 assessTrialStore(88, 237, "totSales")
653 assessTrialStore(88, 237, "nCustomers")
654
655 # Combine
656 trial_results <- rbind(
657     assessTrialStore(77, 233, "totSales"),
658     assessTrialStore(77, 233, "nCustomers"),
659     assessTrialStore(86, 155, "totSales"),
660     assessTrialStore(86, 155, "nCustomers"),
661     assessTrialStore(88, 237, "totSales"),
662     assessTrialStore(88, 237, "nCustomers")
663 )
664
665 trial_results
666
667 # Transactions Per Customer Analysis
668
669 assessTrialStore(77, 233, "nTxnPerCust")
670 assessTrialStore(86, 155, "nTxnPerCust")
671 assessTrialStore(88, 237, "nTxnPerCust")
672
673 txn_results <- rbind(
674     assessTrialStore(77, 233, "nTxnPerCust"),
675     assessTrialStore(86, 155, "nTxnPerCust"),
676     assessTrialStore(88, 237, "nTxnPerCust")
677 )
678
679 txn_results
680
681
682 # Confidence Interval Charts
683
684 createTrialChart <- function(
685     trial_store,
686     control_store,
687     metric_name
688 ) {

```

```

689
690 scalingFactor <- preTrialMeasures[
691   STORE_NBR == trial_store,
692   sum(get(metric_name))
693 ] /
694 preTrialMeasures[
695   STORE_NBR == control_store,
696   sum(get(metric_name))
697 ]
698
699 controlData <- measureOverTime[
700   STORE_NBR == control_store
701 ]
702   , scaledMetric := get(metric_name) * scalingFactor
703 ]
704
705 trialData <- measureOverTime[
706   STORE_NBR == trial_store
707 ]
708
709 comparison <- merge(
710   trialData[, .(YEARMONTH,
711     trialMetric = get(metric_name))],
712   controlData[, .(YEARMONTH,
713     scaledMetric)],
714   by = "YEARMONTH"
715 )
716
717 preTrialDiff <- comparison[
718   YEARMONTH < 201902,
719   abs(trialMetric - scaledMetric) /
720   scaledMetric
721 ]
722
723 stdDev <- sd(preTrialDiff)
724
725 comparison[
726   ,
727   upperCI := scaledMetric * (1 + 2 * stdDev)
728 ]
729
730 comparison[
731   ,
732   lowerCI := scaledMetric * (1 - 2 * stdDev)
733 ]
734
735 comparison[
736   ,
737   Month := as.Date(
738     paste(
739       YEARMONTH %% 100,
740       YEARMONTH %% 100,
741       "01",
742       sep = "-"

```

```

743     )
744   )
745 ]
746
747   return(comparison)
748 }
749
750 chart77 <- createTrialChart(
751   77,
752   233,
753   "totSales"
754 )
755
756 head(chart77)
757
758
759 # Conclusion
760 # Store 77 Insights
761 # Store 77 recorded significant sales growth during March and April 2019.
762 # Customer numbers increased significantly during the trial period.
763 # Transactions per customer also increased significantly.
764 #
765 # The uplift appears to be driven by both increased customer traffic
766 # and increased purchasing behaviour.
767 #
768 # The trial layout was successful and should be considered for rollout.
769
770
771
772
773 ## Store 86 Insights
774 # Store 86 experienced significant sales uplift during February
775 # and March 2019.
776 #
777 # Customer numbers increased significantly throughout the trial period,
778 # while transactions per customer remained unchanged.
779 #
780 # This suggests that the trial layout attracted additional customers
781 # rather than increasing purchase frequency.
782 #
783 # The trial layout appears successful and should be considered
784 # for rollout.
785
786
787 # Store 88 Insights
788 # Store 88 recorded significant sales growth during March and April 2019.
789 #
790 # Customer numbers increased significantly during the same period.
791 # Transactions per customer also increased significantly.
792 #
793 # The trial layout appears to have positively influenced both
794 # customer traffic and purchasing behaviour.
795 #
796 # The trial layout was successful and should be considered

```

```

797 # for rollout.
798
799
800
801
802 # Recommendations
803
804 # Final Recommendation
805 #
806 # The trial layouts implemented in Stores 77, 86 and 88
807 # delivered positive results during the trial period.
808 #
809 # Store 77 demonstrated significant increases in sales,
810 # customer numbers and transactions per customer.
811 #
812 # Store 86 showed strong customer growth and sales uplift,
813 # suggesting the trial successfully attracted additional shoppers.
814 #
815 # Store 88 recorded significant increases in sales,
816 # customer numbers and purchasing frequency.
817 #
818 # Based on the statistical analysis, all three trial stores
819 # outperformed their respective control stores during the trial period.
820 #
821 # It is recommended that the new layout be considered
822 # for broader implementation across suitable stores.
823
824
825
826 # Github Charts
827
828 dir.create("charts/task2", recursive = TRUE, showWarnings = FALSE)
829 # Create reusable GitHub chart function
830
831 createGithubTrialChart <- function(trial_store, control_store, metric_name,
832 chart_title, y_label, file_name) {
833
834     chartData <- createTrialChart(trial_store, control_store, metric_name)
835
836     chartLong <- melt(
837         chartData,
838         id.vars = c("YEARMONTH", "Month"),
839         measure.vars = c("trialMetric", "scaledMetric", "upperCI", "lowerCI"),
840         variable.name = "MetricType",
841         value.name = "Value"
842     )
843
844     chartLong[, MetricType := fifelse(
845         MetricType == "trialMetric", "Trial Store",
846         fifelse(
847             MetricType == "scaledMetric", "Control Store (Scaled)",
848             fifelse(
849                 MetricType == "upperCI", "Upper Confidence Interval",

```

```

850     )
851   )
852 ]]
853
854 p <- ggplot(chartLong, aes(x = Month, y = Value, color = MetricType)) +
855   geom_rect(
856     data = chartData[YEARMONTH >= 201902 & YEARMONTH <= 201904],
857     aes(xmin = min(Month), xmax = max(Month), ymin = -Inf, ymax = Inf),
858     inherit.aes = FALSE,
859     alpha = 0.15
860   ) +
861   geom_line(size = 1) +
862   geom_point(size = 2) +
863   labs(
864     title = chart_title,
865     subtitle = paste("Trial Store", trial_store, "vs Control Store", control_store),
866     x = "Month",
867     y = y_label,
868     color = "Legend",
869     caption = "Shaded area represents trial period: Feb 2019 to Apr 2019"
870   ) +
871   theme_bw() +
872   theme(
873     plot.title = element_text(face = "bold", size = 14),
874     plot.subtitle = element_text(size = 11),
875     axis.text.x = element_text(angle = 45, hjust = 1),
876     legend.position = "bottom"
877   )
878
879 ggsave(
880   filename = paste0("charts/task2/", file_name),
881   plot = p,
882   width = 10,
883   height = 6,
884   dpi = 300
885 )
886
887 return(p)
888 }
889 # Sales Chart
890 createGithubTrialChart(
891   77, 233, "totSales",
892   "Sales Uplift Assessment - Store 77",
893   "Total Sales",
894   "store_77_sales_uplift.png"
895 )
896
897 createGithubTrialChart(
898   86, 155, "totSales",
899   "Sales Uplift Assessment - Store 86",
900   "Total Sales",
901   "store_86_sales_uplift.png"
902 )
903

```

```
904 createGithubTrialChart(  
905     88, 237, "totSales",  
906     "Sales Uplift Assessment - Store 88",  
907     "Total Sales",  
908     "store_88_sales_uplift.png"  
909 )  
910  
911 # Customer Charts  
912  
913 createGithubTrialChart(  
914     77, 233, "nCustomers",  
915     "Customer Uplift Assessment - Store 77",  
916     "Number of Customers",  
917     "store_77_customer_uplift.png"  
918 )  
919  
920 createGithubTrialChart(  
921     86, 155, "nCustomers",  
922     "Customer Uplift Assessment - Store 86",  
923     "Number of Customers",  
924     "store_86_customer_uplift.png"  
925 )  
926  
927 createGithubTrialChart(  
928     88, 237, "nCustomers",  
929     "Customer Uplift Assessment - Store 88",  
930     "Number of Customers",  
931     "store_88_customer_uplift.png"  
932 )  
933  
934 # Transactions per customer charts  
935 createGithubTrialChart(  
936     77, 233, "nTxnPerCust",  
937     "Transaction Frequency Assessment - Store 77",  
938     "Transactions per Customer",  
939     "store_77_transactions_per_customer.png"  
940 )  
941  
942 createGithubTrialChart(  
943     86, 155, "nTxnPerCust",  
944     "Transaction Frequency Assessment - Store 86",  
945     "Transactions per Customer",  
946     "store_86_transactions_per_customer.png"  
947 )  
948  
949 createGithubTrialChart(  
950     88, 237, "nTxnPerCust",  
951     "Transaction Frequency Assessment - Store 88",  
952     "Transactions per Customer",  
953     "store_88_transactions_per_customer.png"  
954 )
```